

SUNNY SHAH

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Education

University of Michigan

Expected: May 2027

Bachelor of Science in Computer Science and Pure Mathematics

Ann Arbor, MI

Relevant Coursework: Data Structures and Algorithms, Computer Science Theory, Discrete Math, Machine Learning, Computer Organization, Probability Theory, Object-Oriented Design, Operating Systems, Networking

Experience

Software Engineer Intern

May 2026 - Present

Capital One

Richmond, VA

- Distributed Systems - AI Innovation Project

Data & Artificial Intelligence Engineering Intern

May 2025 – August 2025

Protiviti

Chicago, IL

- Architected a Streamlit vendor-spend platform that ingests ZIP/CSV uploads, validates schemas, and outputs clean model-ready datasets, driving real-time FP&A insights for executives across Fortune 500 client teams
- Engineered feature pipelines for Random-Forest classification, Prophet forecasting, and Isolation-Forest anomaly detection, producing fully numeric training matrices in under two seconds on commodity hardware for 100k+ rows
- Trained and serialized production-grade models (500-tree RF, tuned Prophet, 300-tree IF) achieving 90% category accuracy and sub-10% 12-month forecast MAPE on validation data while enabling seamless CI/CD deployment
- Optimized ingestion throughput 3× via PyArrow parsing and session-state caching while maintaining minimal memory footprint, ensuring smooth performance on enterprise-scale datasets in cloud-hosted VDI environments

Software Engineer Intern - Research

November 2023 – August 2024

Wayne State University - College of Engineering

Detroit, MI

- Developed a drone design in CAD software, improving lift-induced drag and air resistance by conducting iterative tests and simulations, adjusting wing sizes in NVIDIA Omniverse to accurately model and evaluate real-world performance
- Authored research proposal, outlining project goals, methodology, and expected outcomes, secured funding over \$1,000
- Organized and analyzed over 20 large datasets by implementing advanced NumPy techniques, including vectorized operations, multidimensional array manipulations, and statistical analyses, significantly reducing processing time

Projects

Fault-Tolerant Key-Value Store | *Go, RPC, Distributed Systems*

February 2026

- Implemented a fault-tolerant key-value store using primary/backup replication, with a centralized view service for leader election, failure detection via heartbeat (Ping) protocol, and automated view transitions to safely promote backups
- Built an RPC-based client-server architecture supporting Get, Put, and Append operations, ensuring linearizability and exactly-once semantics through operation deduplication and retry handling, while coordinating primary-to-backup state

Multi-Threaded Network File Server | *C++20, TCP Sockets, Boost Threads, RAII*

December 2025

- Built a multi-threaded TCP file server in C++ using POSIX sockets, Boost threads, and RAII, servicing concurrent RPC-style client requests (Read, Write, Create, Delete) while maintaining strict filesystem consistency guarantees
- Designed a fine-grained concurrency model using reader-writer locks, upgradeable shared locks, and hand-over-hand locking using smart pointers, enabling safe parallel reads/writes while eliminating race conditions, deadlocks, and invalid inode/block references across thousands of concurrent operations

Virtual Memory Pager | *C++20, Operating Systems, Data Structures*

November 2025

- Designed and implemented a demand-paged virtual memory subsystem handling page faults, page tables, swap space, and file-backed mappings, supporting fork with copy-on-write semantics and shared pages across processes
- Optimized pager performance by deferring work and minimizing disk I/O, leveraging copy-on-write, dirty/reference bits, and clock-based eviction, reducing unnecessary page copies and ensuring predictable memory availability under pressure

User-Level Thread Library | *C++20, ucontext, Mutexes, Condition Variables*

October 2025

- Built a preemptive user-level threading library supporting thread creation, join, mutexes, and condition variables, implementing FIFO scheduling, context switching via swapcontext, and monitor semantics across multi-CPU systems
- Implemented low-level synchronization primitives including spin locks (test-and-set), interrupt enable/disable logic, and inter-processor interrupts, ensuring atomicity, preventing missed wakeups, and minimizing unnecessary context switches

Technical Skills

Languages: C++, Python, SQL, JavaScript, HTML/CSS, LaTeX, Bash Scripting

Frameworks: React, Bootstrap, jQuery, PyTorch, OpenCV, Agile, Jira

Developer Tools: Google Cloud Platform (GCP), Amazon Web Services (AWS), Git, VS Code, Jupyter Notebook, Linux

Libraries: pandas, NumPy, Matplotlib, PyTorch, SciPy, Scikit-learn